

# Arden Monaghan

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## EDUCATION

### University of Alberta

Sep. 2022 – Dec 2026

*Bachelor of Science in Computer Science, Specialization*

*Edmonton, AB*

- Cumulative GPA: **3.6** — Major GPA: **3.8**
- **Relevant Coursework:** Data Structures and Algorithms, Introduction to Software Development, Machine Learning 1, Databases & SQL, Statistics 1 & 2, Computer Organization and Architecture, C Programming.
- **Relevant Awards:** Deans Research Award 2025, Jason Lang Scholarship, Deans Honor Roll x2

## RELEVANT EXPERIENCE

### Research Assistant

Jan 2025 – Present

*University of Alberta Electrical and Computer Engineering Faculty*

*Edmonton, AB*

- Implemented a **Computer Vision pipeline in OpenCV** to calibrate multiple cameras and estimate **3D hand-joint positions** using ArUco Marker detection, camera-pose estimation, and frame transformations.
- Integrated **SAM technology** using **PyTorch** to correctly compare 3D hand meshes to ground truth 2D masks. Fine-tuned model on **manually created dataset** of hand-object representations.

### Front-End Developer

Nov 2024 – Present

*LitLife AI Co.*

*Edmonton, AB*

- Managed the integration of FlutterFlow widgets with Supabase backend, ensuring **accurate data mapping** and **secure storage** for consistent application performance.
- **Revised user authorization** using **row-level security** systems to reduced unauthorized access attempts and harmful injections by **30%**.

### Secretary

Apr 2024 – Present

*Undergraduate Artificial Intelligence Society*

*Edmonton, AB*

- **Leading** partnership with the **UA Innovation Fund** to assist in due diligence on **AI-based startups**, evaluating their technological innovation, market potential, and investment viability.
- Guiding over **100 students** per semester in understanding the fundamentals of data cleaning and supervised learning through scikit-learn workshops and seminars.
- **Coordinating** guest speaker sessions **with professors and industry**, creating an environment where experts can share valuable insights for students pursuing AI.

## PROJECTS

### **MentorMatch UAlberta** | Python, Quart, PyQt6, Docker, Ollama

Apr 2024 - Sept 2024

- Automated the task of matching Mentors and Mentees using an **LLM-integrated algorithm** for the **Science Mentorship Program** at UAlberta, reducing time spent matching from **10 days to 10 minutes**.
- Delivered a PyQt GUI desktop app that integrated Llama3 embeddings with Mentor matching criteria. Enhanced overall program **operations by over 50%** while allowing **student data to remain private**.
- **Managed** a team of 4 developers and collaborated with science faculty teams to understand proper **data safety practices** and program requirements.

### **NHL Positivity Index** | Python, HuggingFace, LoRA, Seaborn

Oct 2023 - Jan 2024

- Conducted **Sentiment Analysis** techniques on NHL subreddit data using Praw API to determine the positive sentiment scores **on all 32 NHL teams**.
- **Fine-tuned** cardiffnlp/twitter-roberta-base-sentiment-latest with **5000 manually labeled** hockey comments to create Chelberta Model. Achieved an **accuracy score of 81.2**, an improvement from the base model from 79.2.
- Featured on hockey platforms media sites including CBC radio, OilersNation, and various other outlets.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, C, C++, Dart

**Technologies/Frameworks:** SQL, MongoDB, React.js, Quart, Qt, Supabase, FlutterFlow, PyTorch, OpenCV

**Developer Tools:** Git, Docker, AWS